

Uniform Poincaré inequalities for the discrete de Rham complex of differential forms

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Outline

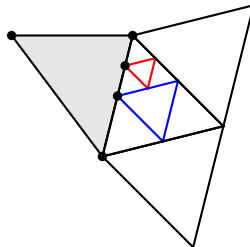
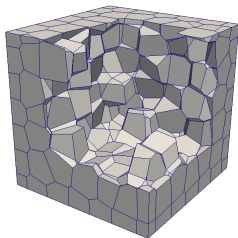
- 1 Discrete polytopal complexes
- 2 Blueprint to establish Poincaré inequality
- 3 Applying the abstract framework
- 4 Extension to differential forms

Discrete polytopal complexes

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Domain and polytopal mesh

- $\Omega \subset \mathbb{R}^n$ polyhedral domain
- $\mathcal{M}_h$ **polyhedral mesh** on Ω



Discrete de Rham method

DDR sequence for $n = 3$:

$$\begin{array}{ccccccc}
 H^1(\Omega) & \xrightarrow{\text{grad}} & \mathbf{H}(\text{curl}; \Omega) & \xrightarrow{\text{curl}} & \mathbf{H}(\text{div}; \Omega) & \xrightarrow{\text{div}} & L^2(\Omega) \\
 \downarrow P_{\text{grad},h} & & \downarrow P_{\text{curl},h} & & \downarrow P_{\text{div},h} & & \downarrow \pi_{\mathcal{P}^k,h} \\
 \underline{X}_{\text{grad},h}^k & \xrightarrow{\underline{G}_h^k} & \underline{X}_{\text{curl},h}^k & \xrightarrow{\underline{C}_h^k} & \underline{X}_{\text{div},h}^k & \xrightarrow{D_h^k} & \mathcal{P}^k(\mathcal{T}_h)
 \end{array}$$

- L^2 -like inner product $(\cdot, \cdot)_{k,h}$

Space	$\mathcal{V}_h$	$\mathcal{E}_h$	$\mathcal{F}_h$	$\mathcal{T}_h$
$\underline{X}_{\text{grad},h}^k$	$\mathbb{R}$	$\mathcal{P}^{k-1}(e)$	$\mathcal{P}^{k-1}(f)$	$\mathcal{P}^{k-1}(t)$
$\underline{X}_{\text{curl},h}^k$		$\mathcal{P}^k(e)$	$\mathcal{RT}^k(f)$	$\mathcal{RT}^k(t)$
$\underline{X}_{\text{div},h}^k$			$\mathcal{P}^k(f)$	$\mathcal{N}^k(t)$
$\mathcal{P}^k(\mathcal{T}_h)$				$\mathcal{P}^k(t)$

Discrete de Rham method

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 \end{array}$$

- Discrete complex property: $\underline{C}_h^k \circ \underline{G}_h^k = 0$, $D_h^k \circ \underline{C}_h^k = 0$

Lemma (Cohomology of the Discrete de Rham complex)

The cohomology of the DDR complex is isomorphic to the cohomology of the continuous de Rham complex, i.e.,

$$\ker \underline{C}_h^k / \text{Im } \underline{G}_h^k \cong \text{Ker } \mathbf{curl} / \text{Im } \mathbf{grad}$$

$$\text{Ker } D_h^k / \text{Im } \underline{C}_h^k \cong \text{Ker } \text{div} / \text{Im } \mathbf{curl}$$

The missing piece: Poincaré inequality

Curl case:

$$\|\underline{\mathbf{v}}_h\|_{\text{curl},h} \lesssim \|\underline{\mathbf{C}}_h^k \underline{\mathbf{v}}_h\|_{\text{div},h} \text{ for all } \underline{\mathbf{v}}_h \in (\text{Ker } \underline{\mathbf{C}}_h^k)^\perp$$

Key point: the constant appearing in the Poincaré inequality does not have to depend on the diameter h of the mesh $\mathcal{M}_h$

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Key point: the constant appearing in the Poincaré inequality does not have to depend on the diameter h of the mesh $\mathcal{M}_h$

- We will focus on the following **equivalent form of Poincaré inequality**:

$$\exists \underline{\mathbf{u}}_h \in \underline{\mathbf{X}}_{\text{curl},h}^k \mid \underline{\mathbf{C}}_h^k \underline{\mathbf{u}}_h = \underline{\mathbf{C}}_h^k \underline{\mathbf{v}}_h \text{ and } \|\underline{\mathbf{u}}_h\|_{\text{curl},h} \lesssim \|\underline{\mathbf{C}}_h^k \underline{\mathbf{v}}_h\|_{\text{div},h}$$

Blueprint to establish Poincaré inequality

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Abstract setting for Poincaré inequalities

Pair of chain complexes (X^k, d^k) and $(\tilde{X}^k, \tilde{d}^k)$ connected via graded maps $(I^k)_k$ and $(R^k)_k$:

$$\begin{array}{ccccccc} \dots & \longrightarrow & X^k & \xrightarrow{d^k} & X^{k+1} & \longrightarrow & \dots \\ & & \uparrow R^k & & \uparrow R^{k+1} & & \\ & & \downarrow I^k & & \downarrow I^{k+1} & & \\ \dots & \longrightarrow & \tilde{X}^k & \xrightarrow{\tilde{d}^k} & \tilde{X}^{k+1} & \longrightarrow & \dots \end{array}$$

- Properties of $(I^k)_k$ and $(R^k)_k$:

- 1 $(I^k)_k, (R^k)_k$ are **cochains maps**, i.e., $\tilde{d}^k \circ I^k = I^{k+1} \circ d^k$,
 $d^k \circ R^k = R^{k+1} \circ \tilde{d}^k$
- 2 I^k, R^k are **bounded** (uniformly in h), i.e., $\|I^k(\cdot)\|_{\tilde{X},k} \lesssim \|\cdot\|_{X,k}$,
 $\|R^k(\cdot)\|_{X,k} \lesssim \|\cdot\|_{\tilde{X},k}$
- 3 $R^k \circ I^k = \text{id}_{X^k}$ on $\text{Im}(d^k)$

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Lemma

If $(\tilde{X}^k, \tilde{d}^k)$ satisfies a Poincaré inequality, then so does (X^k, d^k)

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 & & \uparrow R^k & & \uparrow R^{k+1} & & \\
 & & \downarrow I^k & & \downarrow I^{k+1} & & \\
 \dots & \longrightarrow & \tilde{X}^k & \xrightarrow{\tilde{d}^k} & \tilde{X}^{k+1} & \longrightarrow & \dots
 \end{array}$$

Lemma

If $(\tilde{X}^k, \tilde{d}^k)$ satisfies a Poincaré inequality, then so does (X^k, d^k)

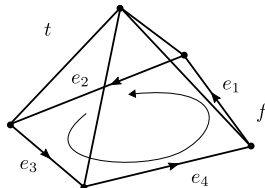
$$\begin{aligned}
 d^k R^k(\tilde{u}) &= R^{k+1} \tilde{d}^k(\tilde{u}) = R^{k+1} \tilde{d}^k I^k(v) = R^{k+1} I^{k+1} d^k(v) = d^k v \\
 \|R^k(\tilde{u})\|_{X,k} &\lesssim \|\tilde{u}\|_{\tilde{X},k} \lesssim \|\tilde{d}^k I^k(v)\|_{\tilde{X},k+1} = \|I^{k+1} d^k(v)\|_{\tilde{X},k+1} \lesssim \|d^k(v)\|_{X,k}
 \end{aligned}$$

Applying the abstract framework to DDR

$$\begin{array}{ccccccc}
 \underline{X}_{\text{grad},h}^k & \xrightarrow{\underline{G}_h^k} & \underline{X}_{\text{curl},h}^k & \xrightarrow{\underline{C}_h^k} & \underline{X}_{\text{div},h}^k & \xrightarrow{D_h^k} & \mathcal{P}^k(\mathcal{T}_h) \\
 \uparrow R_{\text{grad},h} & & \uparrow R_{\text{curl},h} & & \uparrow R_{\text{div},h} & & \uparrow i \\
 & \searrow l_{\text{grad},h} & \searrow l_{\text{curl},h} & & \searrow l_{\text{div},h} & & \searrow \pi_{\mathcal{P}^0,h} \\
 \underline{X}_{\text{grad},h}^0 & \xrightarrow{\underline{G}_h^0} & \underline{X}_{\text{curl},h}^0 & \xrightarrow{\underline{C}_h^0} & \underline{X}_{\text{div},h}^0 & \xrightarrow{D_h^0} & \mathcal{P}^0(\mathcal{T}_h) \\
 \downarrow \kappa_{\text{grad}} \cong & & \downarrow \kappa_{\text{curl}} \cong & & \downarrow \kappa_{\text{div}} \cong & & \downarrow \kappa_{\mathcal{P}} \cong \\
 \mathbb{R}^{\Delta_0(\mathcal{M}_h)} & \xrightarrow{\delta^0} & \mathbb{R}^{\Delta_1(\mathcal{M}_h)} & \xrightarrow{\delta^1} & \mathbb{R}^{\Delta_2(\mathcal{M}_h)} & \xrightarrow{\delta^2} & \mathbb{R}^{\Delta_3(\mathcal{M}_h)}
 \end{array}$$

A closer look to the cochain complex

$$\mathbb{R}\Delta_0(\mathcal{M}_h) \xrightarrow{\delta^0} \mathbb{R}\Delta_1(\mathcal{M}_h) \xrightarrow{\delta^1} \mathbb{R}\Delta_2(\mathcal{M}_h) \xrightarrow{\delta^2} \mathbb{R}\Delta_3(\mathcal{M}_h)$$



- $\delta^0, \delta^1, \delta^2$ play the role of **mimetic operators** and are represented by the **incidence matrices** of the polyhedral mesh $\mathcal{M}_h$, e.g.:

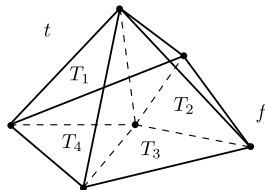
$$\langle \delta^1(\lambda), f \rangle := \sum_{e \in \Delta_1(\mathcal{M}_h)} \textcolor{red}{l_{f,e}} \langle \lambda, e \rangle \quad \textcolor{red}{l_{f,e}} \in \{-1, 0, 1\}$$

- By the very geometric structure of $\mathcal{M}_h$, it holds

$$\delta^1 \delta^0 = 0 \text{ and } \delta^2 \delta^1 = 0$$

A further step

Let $\mathcal{T}_h$ be a **simplicial submesh** of the polyhedral mesh $\mathcal{M}_h$:



$$\begin{array}{ccccccc}
 \mathbb{R}\Delta_0(\mathcal{M}_h) & \xrightarrow{\delta^0} & \mathbb{R}\Delta_1(\mathcal{M}_h) & \xrightarrow{\delta^1} & \mathbb{R}\Delta_2(\mathcal{M}_h) & \xrightarrow{\delta^2} & \mathbb{R}\Delta_3(\mathcal{M}_h) \\
 \begin{array}{c} \uparrow R^0 \\ \downarrow I^0 \end{array} & & \begin{array}{c} \uparrow R^1 \\ \downarrow I^1 \end{array} & & \begin{array}{c} \uparrow R^2 \\ \downarrow I^2 \end{array} & & \begin{array}{c} \uparrow R^3 \\ \downarrow I^3 \end{array} \\
 \mathbb{R}\Delta_0(\mathcal{T}_h) & \xrightarrow{\delta^0} & \mathbb{R}\Delta_1(\mathcal{T}_h) & \xrightarrow{\delta^1} & \mathbb{R}\Delta_2(\mathcal{T}_h) & \xrightarrow{\delta^2} & \mathbb{R}\Delta_3(\mathcal{T}_h) \\
 \downarrow \text{dofs}^{-1}_{\cong} & & \downarrow \text{dofs}^{-1}_{\cong} & & \downarrow \text{dofs}^{-1}_{\cong} & & \downarrow \text{dofs}^{-1}_{\cong} \\
 \mathcal{P}_c^1(\mathcal{T}_h) & \xrightarrow{\text{grad}} & \mathcal{N}^1(\mathcal{T}_h) & \xrightarrow{\text{curl}} & \mathcal{RT}^1(\mathcal{T}_h) & \xrightarrow{\text{div}} & \mathcal{P}^0(\mathcal{T}_h)
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 \end{array}$$

- The bottom sequence satisfies a Poincaré inequality and we can apply our abstract setting

Applying the abstract framework

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Construction of map $I^k : \mathbb{R}^{\Delta_k(\mathcal{M}_h)} \rightarrow \mathbb{R}^{\Delta_k(\mathcal{T}_h)}$

Letting $k \in \{0, \dots, 2\}$, the **cochain map property** requires:

$$\delta^k \circ I^k(\lambda) = I^{k+1} \circ \delta^k(\lambda)$$

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Recursive definition of I^k on the index k

Construction of map $I^k : \mathbb{R}^{\Delta_k(\mathcal{M}_h)} \rightarrow \mathbb{R}^{\Delta_k(\mathcal{T}_h)}$

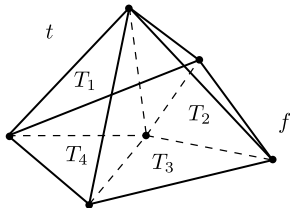
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Recursive definition of I^k on the index k

$\rightsquigarrow$ Fix base case $k = 3$:

$$\langle I^3(\lambda), T_1 \rangle := \frac{|T_1|}{|t|} \langle \lambda, t \rangle$$



Recursive definition of $I^k : \mathbb{R}^{\Delta_k(\mathcal{M}_h)} \rightarrow \mathbb{R}^{\Delta_k(\mathcal{T}_h)}$

- Enforce the cochain map property by taking scalar products on both sides

$$\langle \delta^k \circ I^k(\lambda), w \rangle = \langle I^{k+1} \circ \delta^k(\lambda), w \rangle \quad \forall w \in \mathbb{R}^{\Delta_{k+1}(\mathcal{T}_h)}$$

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- Let ∂_{k+1} be the adjoint of δ^k with respect to the scalar product $\langle \cdot, \cdot \rangle$ and write

$$\langle I^k(\lambda), \partial_{k+1} w \rangle = \langle I^{k+1} \circ \delta^k(\lambda), w \rangle$$

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- Letting $z \in \text{Im } \partial_{k+1}$, $I^k(\lambda)$ is fixed by the expression

$$\langle I^k(\lambda), z \rangle = \langle I^{k+1} \circ \delta^k(\lambda), \partial_{k+1}^{-1} z \rangle$$

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Mesh regularity of $\mathcal{M}_h \rightsquigarrow$

Define I^k by its restriction on each k -cell $c \in \Delta_k(\mathcal{M}_h)$

Recursive definition of $I^k : \mathbb{R}^{\Delta_k(\mathcal{M}_h)} \rightarrow \mathbb{R}^{\Delta_k(\mathcal{T}_h)}$

Lemma

For each k -cell $c \in \Delta_k(\mathcal{M}_h)$ that is not a k -simplex, let

$$Z_k(c, \partial c) := \{z \in \text{Im } \partial_{k+1} \mid \exists S \in \Delta_k(\mathcal{T}_h) \text{ s.t. } \langle S, z \rangle \neq 0 \text{ and } S \not\subset \partial c\}$$

Then, there exist a **basis** $\{z_1, \dots, z_p\}$ of $Z_k(c, \partial c)$ and a set $\{S_1, \dots, S_p\}$ of k -simplexes $S_i \in \Delta_k(\mathcal{T}_h)$ such that:

1

$$S_i \not\subset \partial c \quad \forall i \in \{1, \dots, p\}$$

2

Orthogonality property:

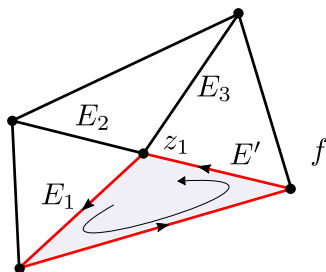
$$\langle S_i, z_j \rangle = \delta_{ij} \quad \forall i, j \in \{1, \dots, p\}$$

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Orthogonality property:

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Example:



Recursive definition of $I^k : \mathbb{R}^{\Delta_k(\mathcal{M}_h)} \rightarrow \mathbb{R}^{\Delta_k(\mathcal{T}_h)}$

- Sum and subtract S in

$$\langle I^k(\lambda), z_i \rangle = \langle I^k(\lambda), S + z_i - S \rangle$$

and recall

$$\langle I^k(\lambda), z_i \rangle = \langle I^{k+1} \circ \delta^k(\lambda), \partial_{k+1}^{-1}(z_i) \rangle$$

to get

$$\langle I^k(\lambda), S \rangle = \langle I^{k+1} \delta^k(\lambda), \partial_{k+1}^{-1}(z_i) \rangle - \langle I^k(\lambda), z_i - S \rangle$$

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- By the orthogonality property $\langle z_i, S_j \rangle = \delta_{ij}$ of the set $\{S_1, \dots, S_p\}$:

$$\text{if } S' \subset c \text{ is s.t. } \langle z_i - S_i, S' \rangle \neq 0 \rightsquigarrow S' \notin \{S_1, \dots, S_p\}$$

and we have two cases:

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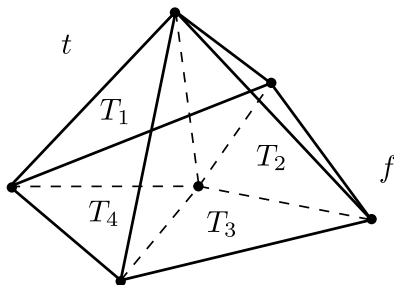
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Recursive definition of I^k on the dimension d

Example: construction of I^k on a pyramidal 3-cell

- Case $k = 3$:

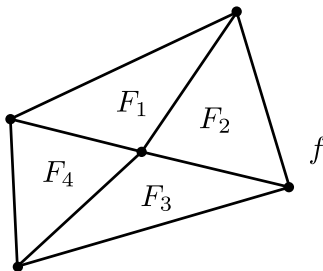
$$\langle I^3(\lambda), T_1 \rangle := \frac{|T_1|}{|t|} \langle \lambda, t \rangle$$



Example: construction of I^k on a pyramidal 3-cell

- Case $k = 2$:
 - Case $d = 2$:

$$\langle I^2(\lambda), F_1 \rangle := \frac{|F_1|}{|f|} \langle \lambda, f \rangle$$

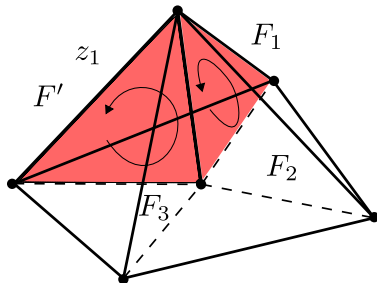


Example: construction of I^k on a pyramidal 3-cell

- Case $k = 2$:

- Case $d = 3$:

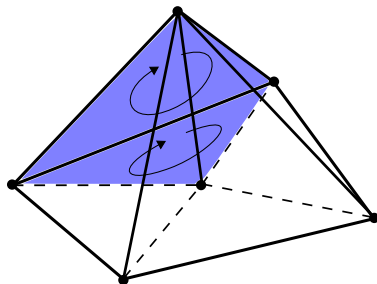
$$\langle I^2(\lambda), F_1 \rangle = \langle I^3 \circ \delta^2(\lambda), \partial_3^{-1}(z_1) \rangle - \langle I^2(\lambda), z_1 - F_1 \rangle$$



Example: construction of I^k on a pyramidal 3-cell

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Example: construction of I^k on a pyramidal 3-cell

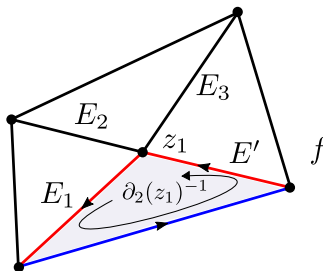
- Case $k = 1$:
 - Case $d = 1$:

$$\langle I^1(\lambda), E \rangle := \frac{|E|}{|e|} \langle \lambda, e \rangle$$

Example: construction of I^k on a pyramidal 3-cell

- Case $k = 1$:
 - Case $d = 2$:

$$\langle I^1(\lambda), E_1 \rangle = \langle I^2 \circ \delta^1(\lambda), \partial_2^{-1}(z_1) \rangle - \langle I^1(\lambda), z_1 - E_1 \rangle$$

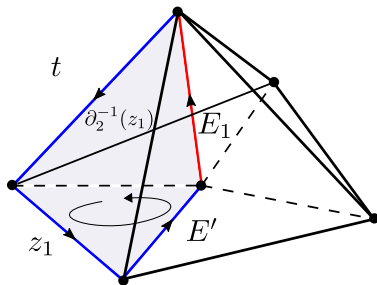


Example: construction of I^k on a pyramidal 3-cell

- Case $k = 1$:

- Case $d = 3$:

$$\langle I^1(\lambda), E_1 \rangle = \langle I^2 \circ \delta^1(\lambda), \partial_2^{-1}(z_1) \rangle - \langle I^1(\lambda), z_1 - E_1 \rangle$$



Example: construction of I^k on a pyramidal 3-cell

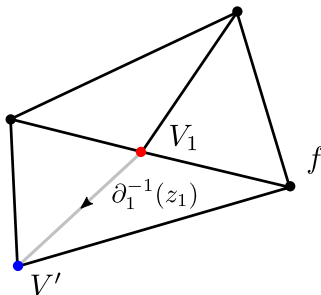
- Case $k = 0$:
 - Case $d = 0$:

$$\langle I^0(\lambda), V \rangle := \langle \lambda, \nu \rangle$$

Example: construction of I^k on a pyramidal 3-cell

- Case $k = 0$:
 - Case $d = 2$:

$$\langle I^0(\lambda), V_1 \rangle = \langle I^1 \circ \delta^0(\lambda), \partial_1^{-1}(z_1) \rangle - \langle I^0(\lambda), z_1 - V_1 \rangle$$

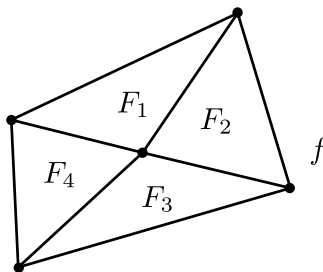


Construction of map $R^k : \mathbb{R}^{\Delta_k(\mathcal{T}_h)} \rightarrow \mathbb{R}^{\Delta_k(\mathcal{M}_h)}$

Easy to define:

$$\langle R^k(\lambda), c \rangle := \sum_{S \in \Delta_k(\mathcal{T}_h), S \subset c} \langle \lambda, S \rangle \quad \lambda \in \mathbb{R}^{\Delta_k(\mathcal{T}_h)}$$

- Example for $k = 2$:

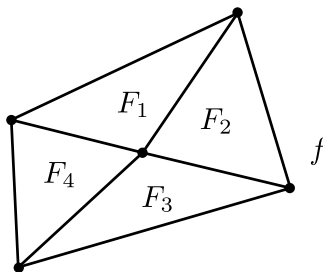


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- Example for $k = 2$:



- R^k is a left inverse of I^k

Summing up

Apply the **abstract Poincaré inequality setting** in two steps

First step:

$$\begin{array}{ccccccc}
 \mathbb{R}\Delta_0(\mathcal{M}_h) & \xrightarrow{\delta^0} & \mathbb{R}\Delta_1(\mathcal{M}_h) & \xrightarrow{\delta^1} & \mathbb{R}\Delta_2(\mathcal{M}_h) & \xrightarrow{\delta^2} & \mathbb{R}\Delta_3(\mathcal{M}_h) \\
 \begin{array}{c} \uparrow R^0 \\ \downarrow I^0 \end{array} & & \begin{array}{c} \uparrow R^1 \\ \downarrow I^1 \end{array} & & \begin{array}{c} \uparrow R^2 \\ \downarrow I^2 \end{array} & & \begin{array}{c} \uparrow R^3 \\ \downarrow I^3 \end{array} \\
 \mathbb{R}\Delta_0(\mathcal{T}_h) & \xrightarrow{\delta^0} & \mathbb{R}\Delta_1(\mathcal{T}_h) & \xrightarrow{\delta^1} & \mathbb{R}\Delta_2(\mathcal{T}_h) & \xrightarrow{\delta^2} & \mathbb{R}\Delta_3(\mathcal{T}_h) \\
 \downarrow \text{dofs} \cong^{-1} & & \downarrow \text{dofs} \cong^{-1} & & \downarrow \text{dofs} \cong^{-1} & & \downarrow \text{dofs} \cong^{-1} \\
 \mathcal{P}_c^1(\mathcal{T}_h) & \xrightarrow{\text{grad}} & \mathcal{N}^1(\mathcal{T}_h) & \xrightarrow{\text{curl}} & \mathcal{RT}^1(\mathcal{T}_h) & \xrightarrow{\text{div}} & \mathcal{P}^0(\mathcal{T}_h)
 \end{array}$$

Summing up

Apply the **abstract Poincaré inequality setting** in two steps

Second step:

$$\begin{array}{ccccccc}
 \underline{X}_{\text{grad},h}^k & \xrightarrow{\underline{G}_h^k} & \underline{X}_{\text{curl},h}^k & \xrightarrow{\underline{C}_h^k} & \underline{X}_{\text{div},h}^k & \xrightarrow{D_h^k} & \mathcal{P}^k(\mathcal{T}_h) \\
 \uparrow R_{\text{grad},h} & & \uparrow R_{\text{curl},h} & & \uparrow R_{\text{div},h} & & \uparrow i \\
 & \searrow I_{\text{grad},h}^0 & \searrow I_{\text{curl},h} & & \searrow I_{\text{div},h} & & \searrow \pi_{\mathcal{P},h}^0 \\
 \underline{X}_{\text{grad},h}^0 & \xrightarrow{\underline{G}_h^0} & \underline{X}_{\text{curl},h}^0 & \xrightarrow{\underline{C}_h^0} & \underline{X}_{\text{div},h}^0 & \xrightarrow{D_h^0} & \mathcal{P}^0(\mathcal{T}_h) \\
 \downarrow \kappa_{\text{grad}} \cong & & \downarrow \kappa_{\text{curl}} \cong & & \downarrow \kappa_{\text{div}} \cong & & \downarrow \kappa_{\mathcal{P}} \cong \\
 \mathbb{R} \Delta_0(\mathcal{M}_h) & \xrightarrow{\delta^0} & \mathbb{R} \Delta_1(\mathcal{M}_h) & \xrightarrow{\delta^1} & \mathbb{R} \Delta_2(\mathcal{M}_h) & \xrightarrow{\delta^2} & \mathbb{R} \Delta_3(\mathcal{M}_h)
 \end{array}$$

Extension to differential forms

- 1 Discrete polytopal complexes
- 2 Blueprint to establish Poincaré inequality
- 3 Applying the abstract framework
- 4 Extension to differential forms

The de Rham complex of differential forms

- **Differential forms:**

- k -forms: mappings ω on Ω s.t. ω_x is a k -alternate linear form on $\mathbb{R}^n$
- d^k : exterior derivative of k -forms
- $H\Lambda^k(\Omega)$: k -forms $\omega \in L^2$ s.t. $d^k\omega \in L^2$

- The de Rham complex with differential forms:

$$H\Lambda^0(\Omega) \xrightarrow{d^0} \dots \xrightarrow{d^{k-1}} H\Lambda^k(\Omega) \xrightarrow{d^k} \dots \xrightarrow{d^{n-1}} H\Lambda^n(\Omega)$$

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$$H\Lambda^0(\Omega) \xrightarrow{d^0} \dots \xrightarrow{d^{k-1}} H\Lambda^k(\Omega) \xrightarrow{d^k} \dots \xrightarrow{d^{n-1}} H\Lambda^n(\Omega)$$

- For $n = 3$, the following links are established through **vector proxies**:

$$\begin{array}{ccccccc} H\Lambda^0(\Omega) & \xrightarrow{d^0} & H\Lambda^1(\Omega) & \xrightarrow{d^1} & H\Lambda^2(\Omega) & \xrightarrow{d^2} & H\Lambda^3(\Omega) \\ \updownarrow & & \updownarrow & & \updownarrow & & \\ H^1(\Omega) & \xrightarrow{\text{grad}} & \mathbf{H}(\text{curl}; \Omega) & \xrightarrow{\text{curl}} & \mathbf{H}(\text{div}; \Omega) & \xrightarrow{\text{div}} & L^2(\Omega) \end{array}$$

Discrete de Rham method with differential forms

DDR sequence of **polynomial differential forms** ⁽¹⁾:

$$\underline{X}_{r,h}^0 \xrightarrow{d_{r,h}^0} \underline{X}_{r,h}^1 \longrightarrow \cdots \longrightarrow \underline{X}_{r,h}^{n-1} \xrightarrow{d_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

- L^2 -like inner product $(\cdot, \cdot)_{k,h}$

Space	$\mathcal{V}_h$	$\mathcal{E}_h$	$\mathcal{F}_h$	$\mathcal{T}_h$
$\underline{X}_{r,h}^0$	$\mathbb{R} = \mathcal{P}_r \Lambda^0(v)$	$\mathcal{P}_r^- \Lambda^1(e)$	$\mathcal{P}_r^- \Lambda^2(f)$	$\mathcal{P}_r^- \Lambda^3(t)$
$\underline{X}_{r,h}^1$		$\mathcal{P}_r \Lambda^0(e)$	$\mathcal{P}_r^- \Lambda^1(f)$	$\mathcal{P}_r^- \Lambda^2(t)$
$\underline{X}_{r,h}^2$			$\mathcal{P}_r \Lambda^0(f)$	$\mathcal{P}_r^- \Lambda^1(t)$
$\underline{X}_{r,h}^3$				$\mathcal{P}_r \Lambda^0(t)$

¹[Bonaldi et al., 2023]

Discrete de Rham method with differential forms

DDR sequence of **polynomial differential forms** ⁽¹⁾:

$$\underline{X}_{r,h}^0 \xrightarrow{\underline{d}_{r,h}^0} \underline{X}_{r,h}^1 \longrightarrow \cdots \longrightarrow \underline{X}_{r,h}^{n-1} \xrightarrow{\underline{d}_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

- Discrete complex property: $\underline{d}_{r,h}^k \circ \underline{d}_{r,h}^{k+1} = 0$

Lemma (Cohomology of the Discrete de Rham complex)

The cohomology of the DDR complex is isomorphic to the cohomology of the continuous de Rham complex, i.e., for all k ,

$$\text{Ker } \underline{d}_{r,h}^k / \text{Im } \underline{d}_{r,h}^{k-1} \cong \text{Ker } d^k / \text{Im } d^{k-1}.$$

¹[Bonaldi et al., 2023]

Poincaré inequality

Strategy: link the DDR complex

$$\underline{X}_{r,h}^0 \xrightarrow{\underline{d}_{r,h}^0} \underline{X}_{r,h}^1 \longrightarrow \cdots \longrightarrow \underline{X}_{r,h}^{n-1} \xrightarrow{\underline{d}_{r,h}^{n-1}} \underline{X}_{r,h}^n$$

with the cochain complex

$$\mathbb{R}\Delta_0(\mathcal{M}_h) \xrightarrow{\delta^0} \cdots \xrightarrow{\delta^{k-1}} \mathbb{R}\Delta_k(\mathcal{M}_h) \xrightarrow{\delta^k} \cdots \xrightarrow{\delta^{n-1}} \mathbb{R}\Delta_n(\mathcal{M}_h)$$

- The construction of the map I^k can be readily applied

Conclusions

- **General proof strategy**: design reduction maps as the canonical embedding of an high-order version into its low-order polynomial space (cf. [Arnold, 2018])
- **Corollary**: in passing, we have proved Poincaré inequalities for low-order mimetic schemes like:
 - **Mimetic Finite Difference** methods [Lipnikov et al., 2014]
 - **Compatible Discrete Operators** [Bonelle, 2014]

Thank you for your attention!



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